

Project Design Phase
Proposed Solution Template

Date	19 September 2026
student ID	23NU1A4429
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Enterprise employees face severe delays (48+ hours) when requesting network access, VPN setup, or firewall rule modifications. The legacy process relies on manual emails and static forms lacking input validation, leading to incomplete submissions, zero status visibility, and high operational overhead for Network Engineers.
2.	Idea / Solution description	An automated, end-to-end Service Catalog solution built on ServiceNow. It utilizes dynamic Catalog UI Policies with IPv4 Regex validation for error-free inputs, Flow Designer for multi tier conditional approvals (Manager / Network Security), auto spawning fulfillment tasks (sc_task), and real-time CMDB target device mapping (cmdb_ci_netgear).
3.	Novelty / Uniqueness	• Zero-Touch Input Validation: Real-time IPv4 format and subnet pattern checks directly on the Service Portal • Contextual Variable Mapping: Technical parameters map directly into engineer task views without back-and-forth communication. • Dynamic One-Click Approvals: In-line email approval actions allowing fast manager response times. • Automated Infrastructure Reconciliation: Real-time updates to CMDB records upon task closure.

4.	Social Impact / Customer Satisfaction	Drastically improves employee productivity by reducing network request fulfillment time from days to hours (8-hour SLA for High/Critical, 24-hour SLA for Standard). Eliminates friction between IT and business teams through transparent, real-time portal tracking and guaranteed SLA resolution windows.
5.	Business Model (Revenue Model)	Internal cost-savings and productivity ROI model. Reduces IT operational labor costs by eliminating manual email exchanges, decreases ticket bounce rates by 90%, and prevents expensive security or audit non-compliance penalties through audit-ready logging (sysapproval_approver).
6.	Scalability of the Solution	Built on ServiceNow's enterprise cloud-native architecture. It can seamlessly scale horizontally to handle thousands of concurrent global requests, expand to support multi-cloud network provisioning (AWS/Azure spokes via IntegrationHub), and adapt to future automated zero-touch firewall API integrations.